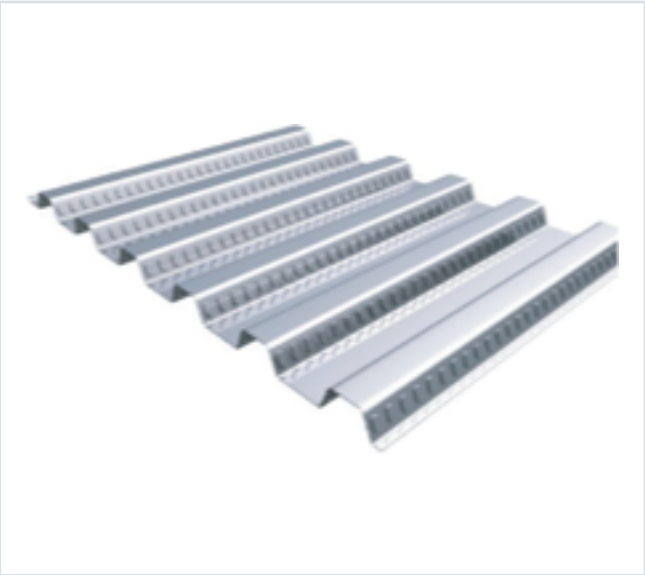


DECKING PRODUCT SUBMITTAL

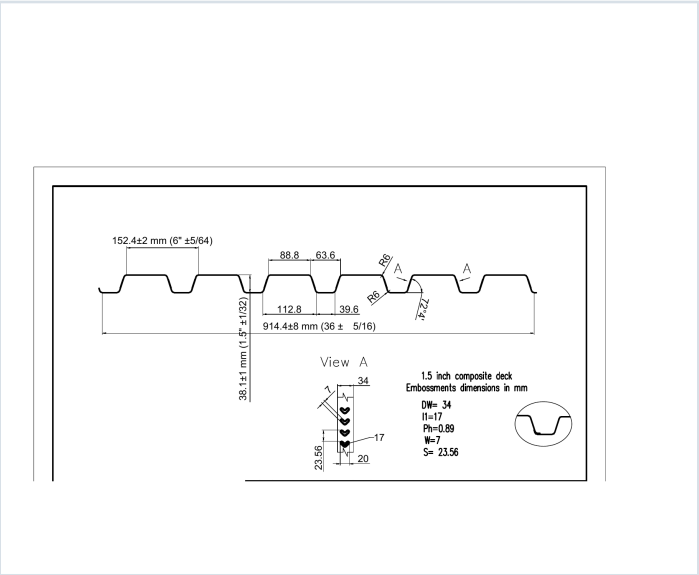
15CD-FY50-20-G90

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 50	20 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	2.0	50	0.216	0.222	6457	6661	0.182	0.210

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 45, 46, 47, 48
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE

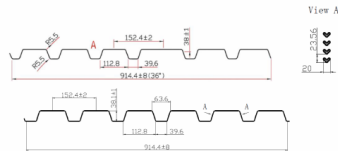
15CD-FY50-20-G90 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 45

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	F _y (ksi)	F _t (ksi)	S _x (in ³)	S _y (in ³)	ASD (S _x + 147)	I _x (in ⁴)	I _y (in ⁴)
22	0.0295	16	50	0.770	0.222	0.001	558	0.144	0.107
20	0.0268	20	50	0.294	0.330	0.001	558	0.182	0.290
18	0.02474	24	50	0.294	0.330	0.001	558	0.257	0.290
16	0.02288	30	50	0.278	0.390	0.001	558	0.341	0.363

Note: All section properties and ASD/bearing strengths are calculated in accordance with ANS/SEI R1-2017, AISI S100-2012 and AISI S100-2018.

Shear and Web Crippling

Gage	V _u /Q (lb/ft)	Web Crippling (R _u /Q, lbs/ft) One Flange Loading End Bearing	Web Crippling (R _u /Q, lbs/ft) Two Flange Loading Interior Bearing
22	2424	1015	1015
20	3803	1535	1535
18	5052	2077	2077
16	6279	2597	2597

Note: All section properties and ASD/bearing strengths are calculated in accordance with ANS/SEI R1-2017, AISI S100-2012 and AISI S100-2018.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	158	12	94	80	69	60	53	47	42	38	34
Double	22	158	12	94	80	69	60	53	47	42	38	34
Triple	22	158	12	94	80	69	60	53	47	42	38	34

STEEL CATALOG PAGE 46

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	16	57	44	35	29	22	18	15	13	11	9
Double	22	16	57	44	35	29	22	18	15	13	11	9
Triple	22	16	57	44	35	29	22	18	15	13	11	9

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table - 20 psf Construction Load

Total Slab Depth	Normal Weight Concrete (145 pcf)	Lightweight Concrete (115 pcf)
3'-0" (1'-0") 31 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
4'-0" (1'-0") 43 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
5'-0" (1'-0") 55 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
6'-0" (1'-0") 67 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
7'-0" (1'-0") 79 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
8'-0" (1'-0") 91 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
9'-0" (1'-0") 103 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
10'-0" (1'-0") 115 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
11'-0" (1'-0") 127 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
12'-0" (1'-0") 139 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
13'-0" (1'-0") 151 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
14'-0" (1'-0") 163 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
15'-0" (1'-0") 175 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
16'-0" (1'-0") 187 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
17'-0" (1'-0") 199 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
18'-0" (1'-0") 211 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
19'-0" (1'-0") 223 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"
20'-0" (1'-0") 235 PSF	15x16 ga 8'-0" 7'-0" 7'-0"	15x16 ga 8'-0" 7'-0" 7'-0"

Note: Web crippling and clear base not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY50-20-G90 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 47

LSC Deck - Wide Flange Up				ISNV Deck - Wide Flange Down			
50 ksi				50 ksi			
Live Load	20	psf	with concrete	Live Load	20	psf	with concrete
DL	160	psf		DL	160	psf	
Deck Weight	2	psf		Deck Weight	2	psf	
Live Load	50	psf	without concrete	Live Load	50	psf	without concrete
Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC	Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC
3.5	22	36	36	3.5	22	25	25
3.5	20	31	31	3.5	20	30	30
3.5	18	40	40	3.5	18	38	38
3.5	16	47	47	3.5	16	46	46
4	22	36	36	4	22	25	25
4	20	31	31	4	20	30	30
4	18	40	40	4	18	38	38
4	16	47	47	4	16	46	46
4.5	22	36	36	4.5	22	25	25
4.5	20	31	31	4.5	20	30	30
4.5	18	40	40	4.5	18	38	38
4.5	16	47	47	4.5	16	46	46
5	22	36	36	5	22	25	25
5	20	31	31	5	20	30	30
5	18	40	40	5	18	38	38
5	16	47	47	5	16	46	46
5.5	22	36	36	5.5	22	25	25
5.5	20	31	31	5.5	20	30	30
5.5	18	40	40	5.5	18	38	38
5.5	16	47	47	5.5	16	46	46
6	22	36	36	6	22	25	25
6	20	31	31	6	20	30	30
6	18	40	40	6	18	38	38
6	16	47	47	6	16	46	46

Assumes that backspan is greater than the cantilever spans.
Applicable to single, double, triple spans or greater.
Controlled by concentrated load = 30 psf
Controlled by concrete weight = 30 psf
Controlled by Deflection

STEEL CATALOG PAGE 48

22 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	372	334	289	252	210	170
4	37	400	400	400	397	340	298	258	218
4.5	43	400	400	400	400	400	397	331	281
5	49	400	400	400	400	400	400	369	319
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	277	156	135	121	110	96	85	80
4	224	197	175	156	139	125	113	102
4.5	273	241	213	190	170	153	138	126
5	323	286	253	226	200	182	164	149
5.5	375	331	294	262	235	211	191	173
6	400	378	336	300	269	242	218	197

20/18/16 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	400	400	400	400	400	400
4	37	400	400	400	400	400	400	400	400
4.5	43	400	400	400	400	400	400	400	400
5	49	400	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	236	181	170	151	136	122	110	100
4	274	242	215	190	172	155	140	127
4.5	335	296	263	235	215	195	172	156
5	398	362	315	280	251	226	203	186
5.5	400	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Note
Due to the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gauge. However, the construction square do get longer for 18 and 16 gauge deck.